Exp. No: 12.
Date:

N-Map to discover live hosts.

Aim:

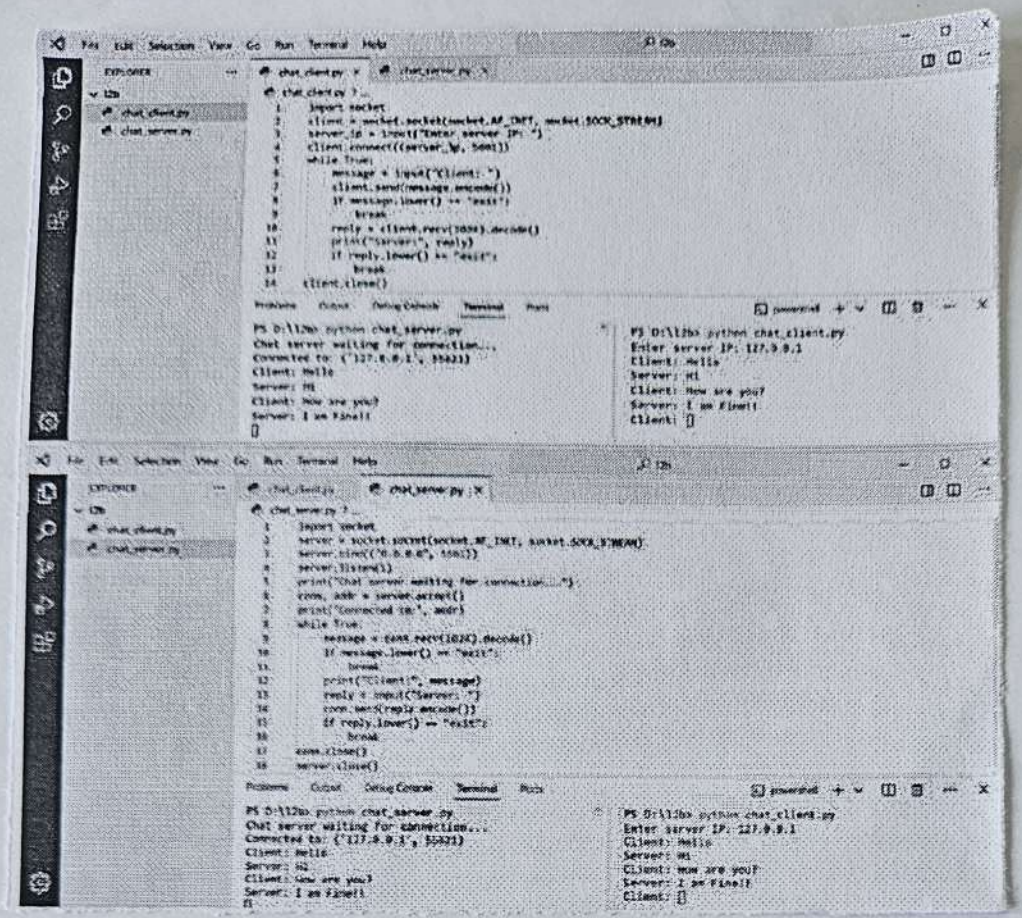
To discover live hosts in a performing ARP scan, ICMP, ping scan, TCP ping scan and UDP ping scan on tryHackme platform.

Procedure:

1. Log in to the tryHackme platform.
2. Open the N-Map room and start the target machine.
3. Start the attachbox and open terminal
4. Enter target machine's IP address.
5. Perform ARP, ICMP and UDP scans using Nmap commands.
6. Observe and record live hosts discovered.

Results:

Thus, live hosts in network discovered successfully using N-Map, with ARP, ICMP, TCP and UDP scanning techniques.



```
import socket
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server_ip = input("Enter server IP: ")
client.connect((server_ip, 5001))
while True:
    message = input("Client: ")
    client.send(message.encode())
    if message.lower() == "exit":
        break
    reply = client.recv(1024).decode()
    print("Server:", reply)
    if reply.lower() == "exit":
        break
client.close()
```

```
PS D:\12b> python chat_server.py
Chat server waiting for connection...
Connected to: ('127.0.0.1', 55421)
Client: Hello
Server: Hi
Client: How are you?
Server: I am Fine!!
```

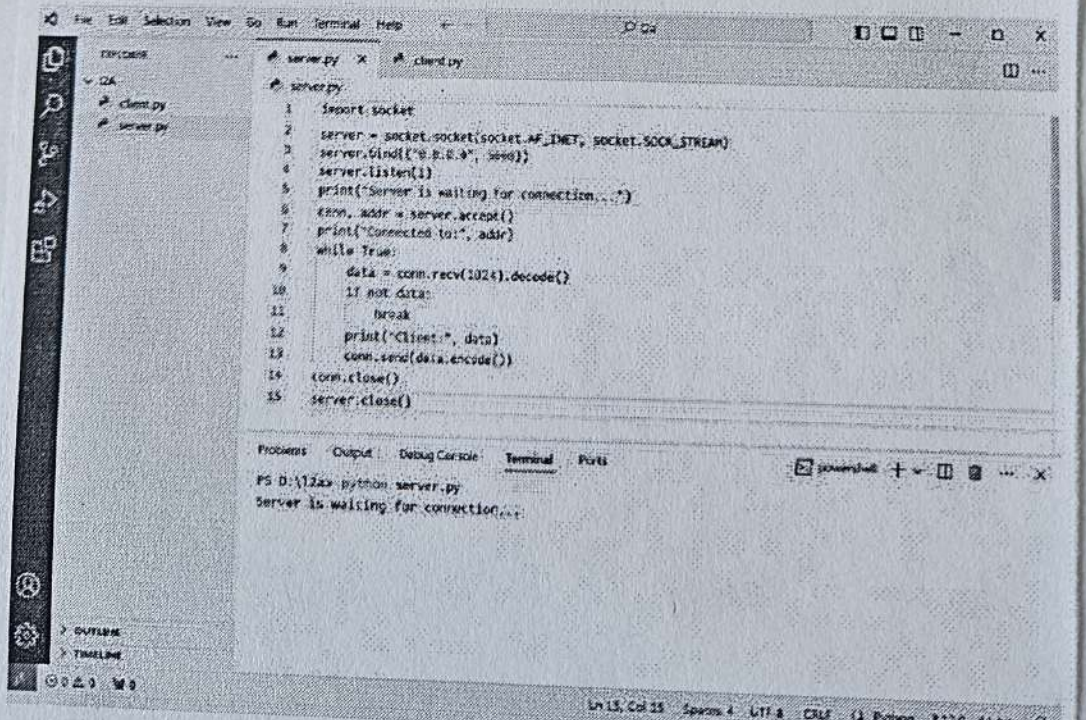
```
PS D:\12b> python chat_client.py
Enter server IP: 127.0.0.1
Client: Hello
Server: Hi
Client: How are you?
Server: I am Fine!!
Client:
```



```
import socket
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind(("0.0.0.0", 5001))
server.listen(1)
print("Chat server waiting for connection...")
conn, addr = server.accept()
print("Connected to:", addr)
while True:
    message = conn.recv(1024).decode()
    if message.lower() == "exit":
        break
    print("Client:", message)
    reply = input("Server: ")
    conn.send(reply.encode())
    if reply.lower() == "exit":
        break
conn.close()
server.close()
```

```
PS D:\12b> python chat_server.py
Chat server waiting for connection...
Connected to: ('127.0.0.1', 55421)
Client: Hello
Server: Hi
Client: How are you?
Server: I am Fine!!
```

```
PS D:\12b> python chat_client.py
Enter server IP: 127.0.0.1
Client: Hello
Server: Hi
Client: How are you?
Server: I am Fine!!
Client:
```



```
import socket
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind(("0.0.0.0", 5000))
server.listen(1)
print("Server is waiting for connection...")
conn, addr = server.accept()
print("Connected to:", addr)
while True:
    data = conn.recv(1024).decode()
    if not data:
        break
    print("Client:", data)
    conn.send(data.encode())
conn.close()
server.close()
```

```
PS D:\12a> python server.py
Server is waiting for connection...
```